Nonexistence of bi-infinite geodesics in exponential last passage percolation

Joint with
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PAD 3-minute talks
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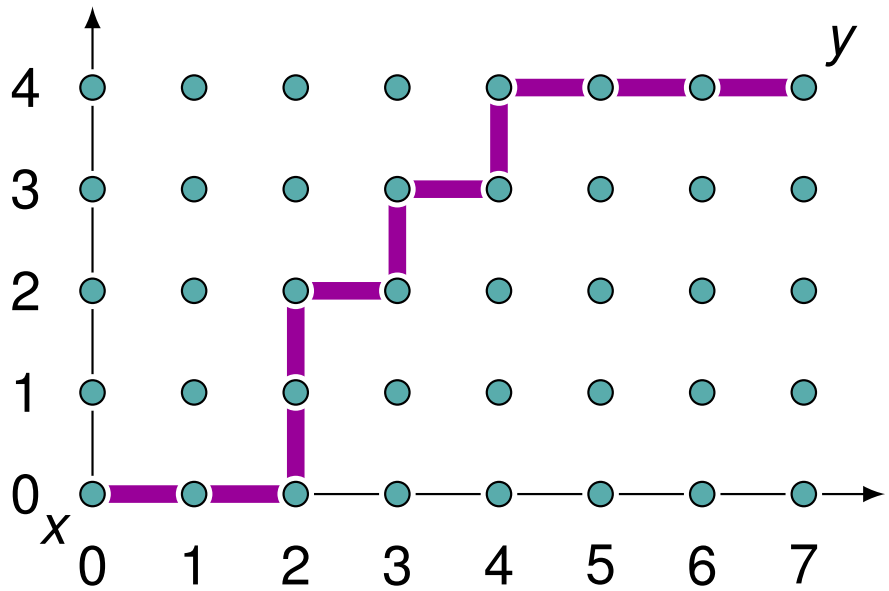
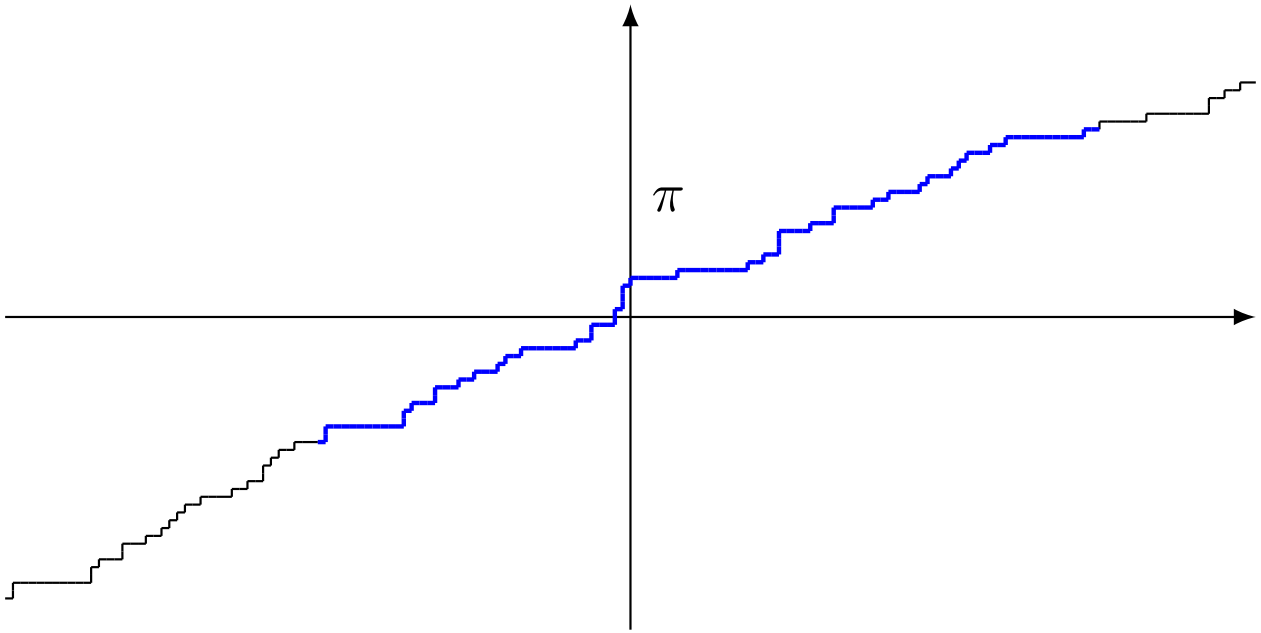
Last passage percolation

- Place ω_z i.i.d. Exp(1) for $z \in \mathbb{Z}^2$.
- The *geodesic* $\pi_{x,y}$ from x to y is the a.s. unique heaviest up-right from x to y.



Bi-infinite geodesics

A bi-infinite up-right path is a *bi-infinite geodesic*, if any of its segments is itself a geodesic between the two endpoints.



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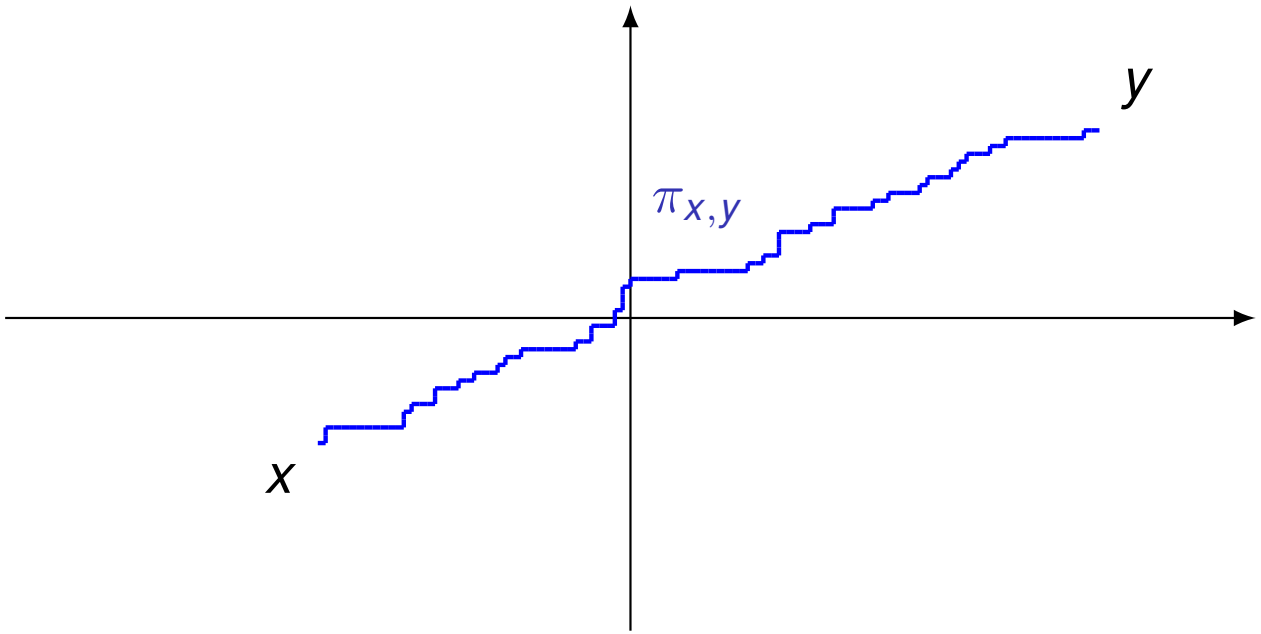
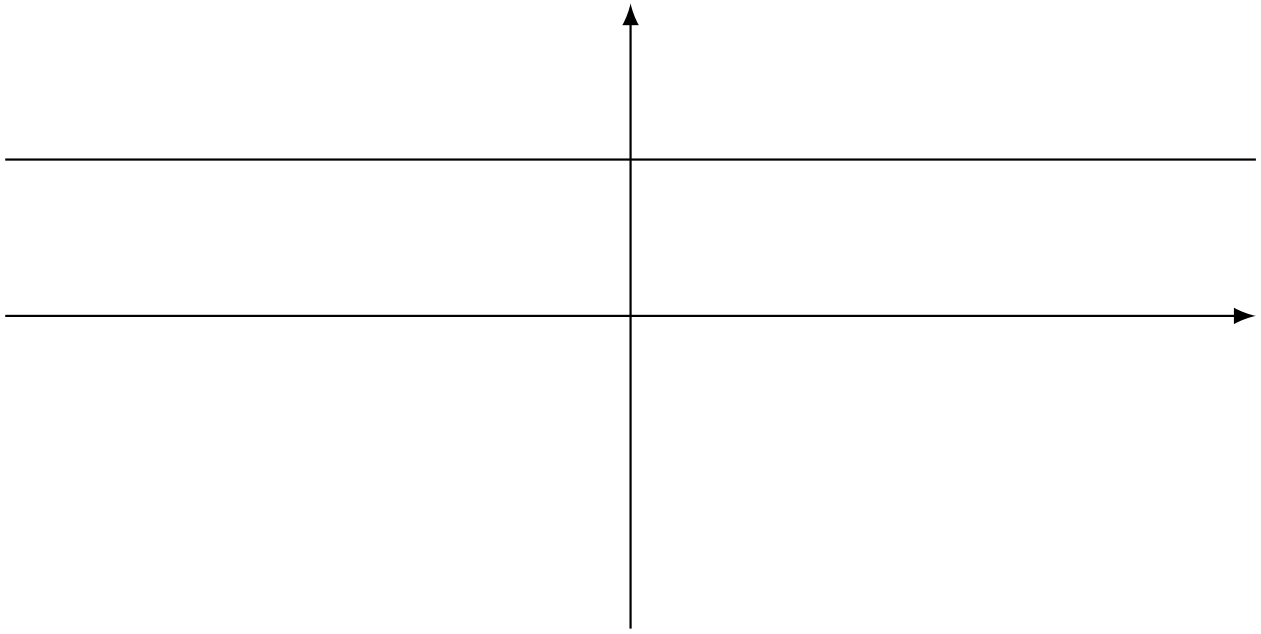


Bi-infinite geodesics

Trivial bi-infinite geodesics:



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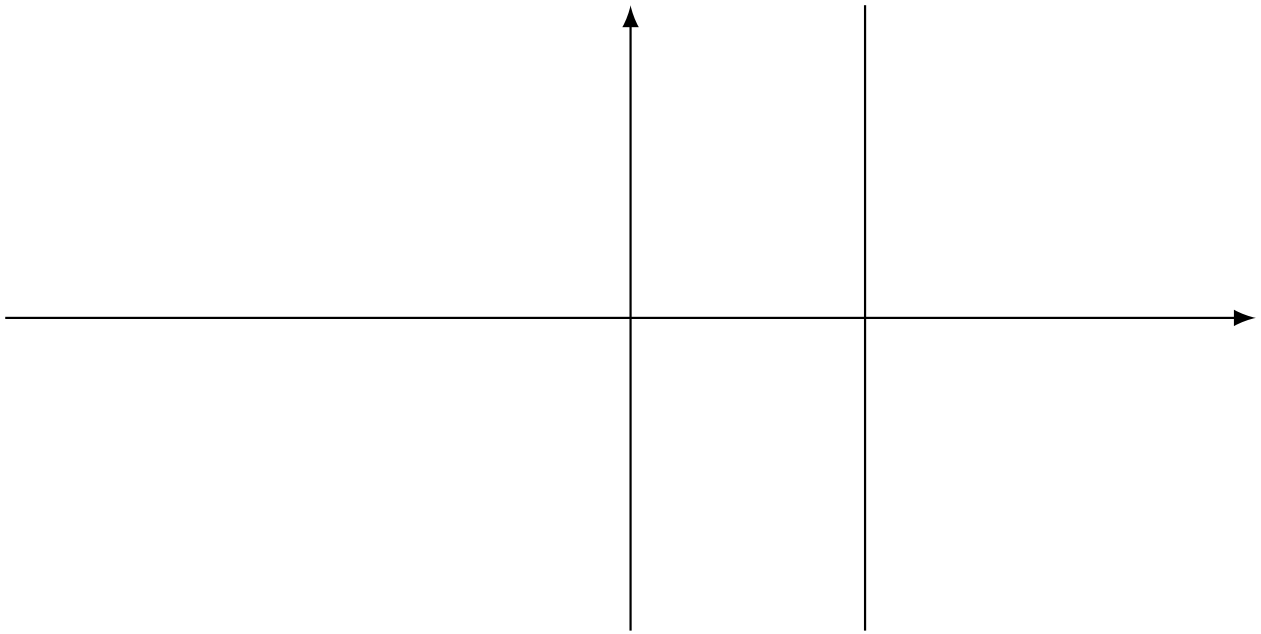

The result

Theorem

A.s., there are no non-trivial bi-infinite geodesics.

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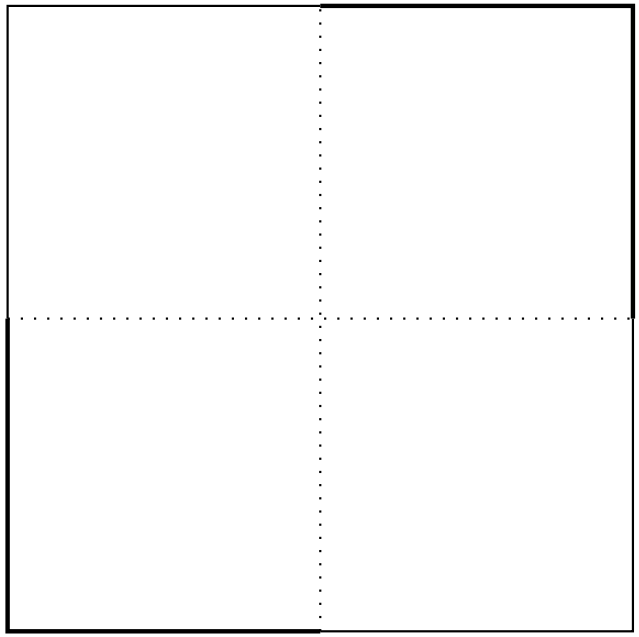
There is too much coalescence going on. The geodesics is going further and further out.

Proof

Take large boxes and show that geodesics avoid the origin when crossing from one side to the other (Newman '95).



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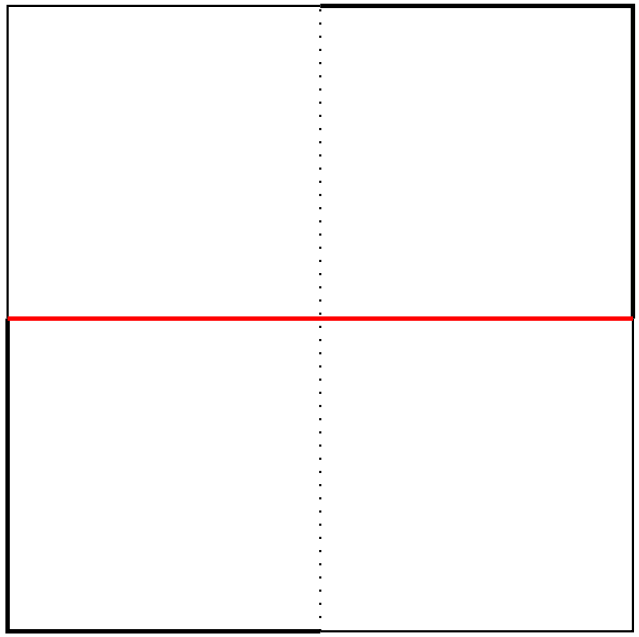
Take large boxes and show that geodesics avoid the origin when crossing from one side to the other (Newman '95).

1. Close to vertical and horizontal geodesics become trivial.



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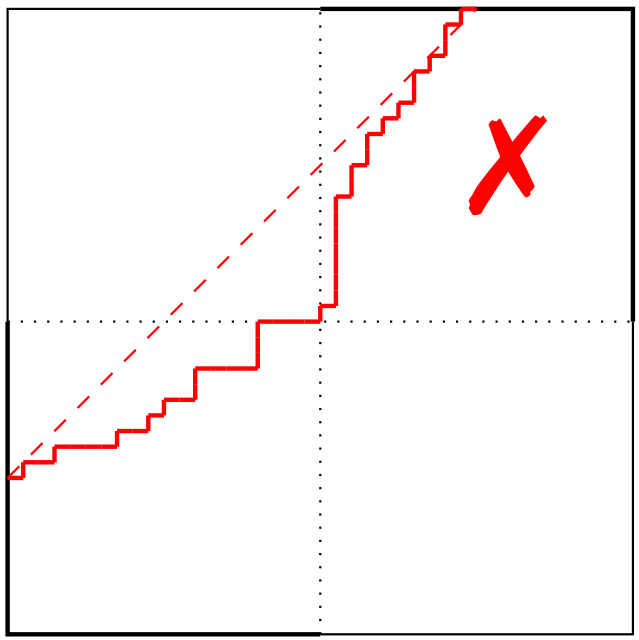
1. Close to vertical and horizontal geodesics become trivial.
2. Otherwise, geodesics don't like to turn too much.



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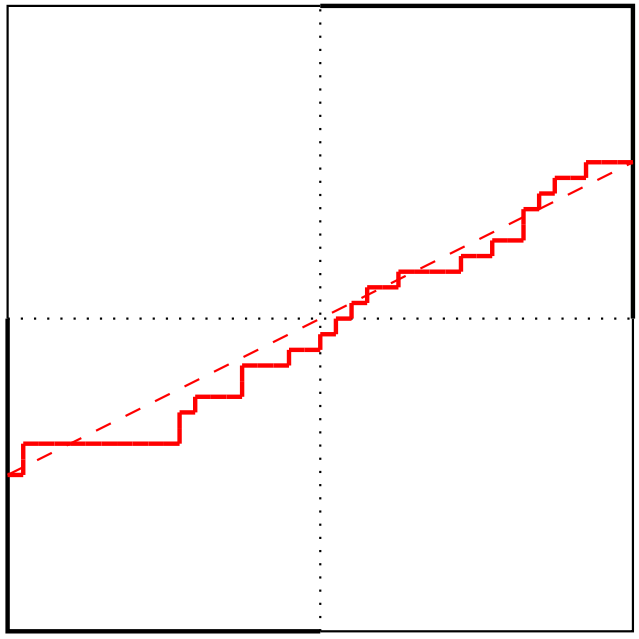
1. Close to vertical and horizontal geodesics become trivial.
2. Otherwise, geodesics don't like to turn too much.
3. We are left with roughly diagonal ones, show that they fluctuate too much.



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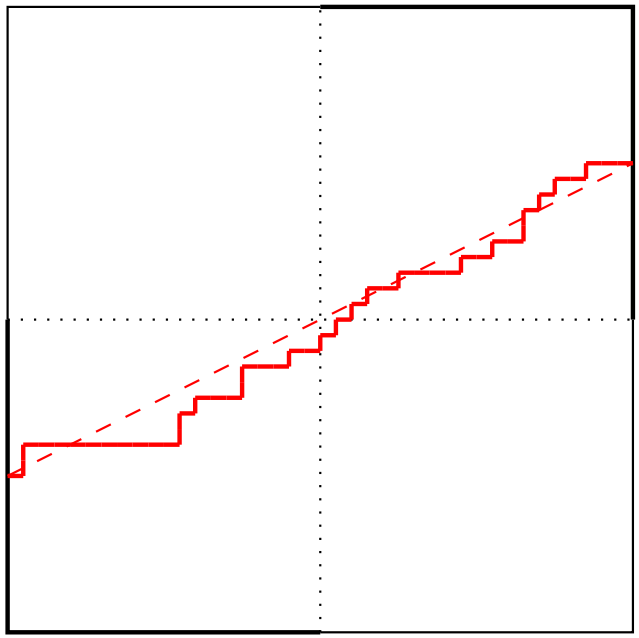

Thank you.